John Muschelli

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Education

2012–2016 **PhD**, *Biostatistics*,
Johns Hopkins Bloomberg School of Public Health, Baltimore, MD.
Computational Methods for Neuroimaging in R: Stroke Hemorrhage in X-ray Computed Tomography Scanning
Advisor: Ciprian Crainiceanu, PhD

2008–2010 **ScM**, *Biostatistics*,
Johns Hopkins Bloomberg School of Public Health, Baltimore, MD.
An Iterative Approach to Hemodynamic Response Function Temporal Derivatives in Statistical Parametric Mapping for Functional Neuroimaging
Advisor: Brian Caffo, PhD

2004–2008 **BS**, *Biomathematics and Neuroscience*,
The University of Scranton, Scranton, PA.
Advisors: Professor Jakub Jasinski, Professor J. Timothy Cannon

Relevant Experience

2022–Present **Associate Research Professor**, *Department of Biostatistics*, Johns Hopkins Bloomberg School of Public Health, (Research-track Faculty).

2020–2022 **Associate Scientist**, *Department of Biostatistics*, Johns Hopkins Bloomberg School of Public Health, (Research-track Faculty).

2016–2020 **Assistant Scientist**, *Department of Biostatistics*, Johns Hopkins Bloomberg School of Public Health, (Research-track Faculty).

2012–2016 **Trainee**, *T32AG021334: Epidemiology and Biostatistics of Aging Training Grant*, Mentors: Dr. Michelle Carlson, Dr. Ravi Varadhan.

2009–2016 **Research Associate**, *Johns Hopkins Biostatistics Consulting Center*, Baltimore, MD.
Collaborated on statistical projects with senior consultants.
Weekly consulting for student research projects.
Report writing and analyzing data using statistical software: R, Stata.

2009–2014 **Data Analyst / Data Manager**, *Brain Injury Outcomes Division*, Baltimore, MD.
Decreased turnaround time on data safety report (from weeks to hours) by using knitr, LaTeX, and dynamic documents.
Created a standardized database and processing pipeline for CT images.
Analyzed phase II and III trials for treatment of intracerebral hemorrhage
Data management and consultation of electronic case report form (eCRF) creation.

2010–2012 **Data Analyst**, *Laboratory for Neurocognitive and Imaging Research at Kennedy Krieger Institute*, Baltimore, MD.
Reduced manual steps in complex imaging study analysis using automation from programming.
Analysis of functional MRI (fMRI) imaging studies using Statistical Parametric Mapping.
Programming consultant: Matlab & R.

Software

R Packages

All download counts are from RStudio CRAN logs and are accurate as of November 06, 2026.

rscopus: Scopus Database API Interface, *625602*.

neurobase: Neuroconductor Base Package with Helper Functions for nifti Objects, *110710*.

diffr: Display Differences Between Two Files using Codediff Library, *92281*.

fslr: Wrapper Functions for FSL (FMRIB Software Library) from Functional MRI of the Brain (FMRIB), *80713*.

gifti: Reads in Neuroimaging GIFTI Files with Geometry Information, *79741*.

brainR: Helper Functions to misc3d and rgl Packages for Brain Imaging, *61943*.

matlabr: An Interface for MATLAB using System Calls, *58580*.

mscstts: R Client for the Microsoft Cognitive Services Text-to-Speech REST API, *52406*.

WhiteStripe: White Matter Normalization for Magnetic Resonance Images, *49313*.

cifti: Toolbox for Connectivity Informatics Technology Initiative (CIFTI) Files, *42259*.

freesurfer: Wrapper Functions for FreeSurfer, *41083*.

gcite: Google Citation Parser, *40272*.

text2speech: Text to Speech Conversion, *39625*.

kirby21.base: Example Data from the Multi-Modal MRI Reproducibility Resource, *38182*.

glassdoor: Interface to Glassdoor API, *37755*.

neurohcp: Human Connectome Project Interface, *37022*.

kirby21.t1: Example T1 Structural Data from the Multi-Modal MRI Reproducibility Resource, *33316*.

kirby21.fmri: Example Functional Imaging Data from the Multi-Modal MRI Reproducibility Resource, *32720*.

stapler: Simultaneous Truth and Performance Level Estimation, *29972*.

spm12r: Wrapper Functions for SPM (Statistical Parametric Mapping) Version 12 from the Wellcome Trust Centre for Neuroimaging, *29630*.

papayar: View Medical Research Images using the Papaya JavaScript Library, *25347*.

leanpubr: Leanpub API Interface, *23526*.

nsrr: Interface to National Sleep Research Resource, *17888*.

fedreporter: Interface to Federal RePORTER API, *15116*.

neurovault: Neurovault Database API Access, *2052*.

GitHub **drammsr: Port of Deformable Registration via Attribute Matching and Mutual-Saliency Weighting (DRAMMS) Registration to R**.

didactr: Tools for Creating Automated Courses.

flexconnr: FLEXCONN Model Wrapped in R .

extrantsr: Additional functionality and extensions to the ANTsR R package.

rcamino: R Port of Camino Software.

dcm2niir: R wrapper for dcm2nii DICOM converter.

ichseg: ICH Segmentation of CT scans.

msseg: MS Lesion Segmentation.

googleCite: Scraper for Google Citations.

processVISION: Scripts for Parsing XML from VISION database.

sri24: SRI24 Atlas: Normal Adult Brain Anatomy.

Shiny Web Applications

2019 **Do Data Analysis with Your Voice!**,
https://jhubiostatistics.shinyapps.io/Speak_dplyr/.

Turning Slide Decks into Videos (private request needed),
https://jhubiostatistics.shinyapps.io/presentation_to_video/.

A Sortable NIH RFA Table,
https://jhubiostatistics.shinyapps.io/rfa_sort/.

Turn a Folder of Slides into a Leanpub Course,
https://jhubiostatistics.shinyapps.io/slides_to_leanpub/.

2016 **Segmentation of Intracranial Hemorrhage from CT Scans**,
http://johnmuschelli.com/ich_segment_all.html.

2015 **Abandoned Cars in Baltimore Finder**,
https://jmuschelli.shinyapps.io/Abandoned_Baltimore_Car.

Unofficial ENAR 2015 Itinerary Maker,
https://muschellij2.shinyapps.io/ENAR_2015.

2014 **Online DICOM TO NIfTI Converter**,
https://muschellij2.shinyapps.io/dcm2nii.

Cost of most common medical procedures at United States hospitals based on Centers for Medicare and Medicaid Services data,
https://jmuschelli.shinyapps.io/Shiny_Health_Data.

Peer-Reviewed Publications

* denotes authors contributed equally

2025 Gao, S., Zhou, X., Koffman, L., Wanigatunga, A. A., Schrack, J. A., Crainiceanu, C. M., **Muschelli, J.,** (2025). “Comparing step counting algorithms for high-resolution wrist accelerometry data in older adults in the ARIC study”. *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences*, glaf034.

Goeddel, L. A., Koffman, L., Hernandez, M., Whitman, G., Parikh, C. R., Lima, J. A., Bandeen-Roche, K., Zhou, X., **Muschelli, J.**, Crainiceanu, C., Faraday, N., Brown IV, C., (2025d). “Occurrence of low cardiac index during normotensive periods in cardiac surgery: a prospective cohort study using continuous noninvasive cardiac output monitoring”. *Anesthesia & Analgesia* 140.1, pp. 77–86.

Goeddel, L. A., Hernandez, M., Koffman, L., Murphy, Z., Khanna, A. K., Robich, M., Whitman, G., Zhou, X., Bandeen-Roche, K., **Muschelli, J.**, Parikh, C. R., Lima, J. A. C., Crainiceanu, C. M., Brown IV, C., Faraday, N., (2025a). “Fine-mapping the association of acute kidney injury with mean arterial and central venous pressures during coronary artery bypass surgery”. *Anesthesia & Analgesia* 140.6, pp. 1439–1449.

Goeddel, L. A., Hernandez, M., Koffman, L., Santino, C., **Muschelli, J.**, Zhou, X., Waldron, N., Parikh, C. R., Lester, L., Abozaid, R., Faraday, N., Crainiceanu, C. M., Brown, C., (2025b). “Assessment of renal vein flow index by transesophageal echocardiography: precision, variability, and association with cardiac index during cardiac surgery”. *Journal of Cardiothoracic and Vascular Anesthesia* 39.9, pp. 2307–2314. ISSN: 1053-0770. DOI: https://doi.org/10.1053/j.jvca.2025.05.052. URL: https://www.sciencedirect.com/science/article/pii/S1053077025004458.

Goeddel, L. A., Hernandez, M., Koffman, L., Slowey, C., **Muschelli, J.**, Zhou, X., Parikh, C. R., Lima, J. A., Bandeen-Roche, K., Faraday, N., Crainiceanu, C. M., Brown IV, C., (2025c). “Assessment of renal vein stasis index by transesophageal echocardiography during cardiac surgery: a feasibility study”. *Anesthesia & Analgesia* 140.1, pp. 224–227.

Koffman, L., Crainiceanu, C., **Muschelli, J.,** (2025). “Comparing step counting algorithms for high-resolution wrist accelerometry data in NHANES 2011-2014”. *Medicine & Science in Sports & Exercise*, pp. 10–1249.

Newsome, S. D., Fitzgerald, K. C., Hughes, A., Beier, M., Koshorek, J., Wang, Y., Maldonado, D. P., Shoemaker, T., Malik, T., Bayu, T., Ravenna, P. E., Avornu, A., Sotirchos, E. S., Romba, M., **Muschelli, J.**, Brown, T. T., Calabresi, P. A., Mowry, E. M., (2025). “Intranasal insulin for improving cognitive function in multiple sclerosis”. *Neurotherapeutics*, e00581. ISSN: 1878-7479. DOI: https://doi.org/10.1016/j.neurot.2025.e00581. URL: https://www.sciencedirect.com/science/article/pii/S1878747925000595.

Zhao, A., Cui, E., Leroux, A., Zhou, X., **Muschelli, J.**, Lindquist, M. A., Crainiceanu, C. M., (2025). “Objectively measured physical activity using wrist-worn accelerometers as a predictor of incident alzheimer’s disease in the uk biobank”. *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences* 80.2, glae287.

2024 Koffman, L., **Muschelli, J.,** (2024). “Evaluating step counting algorithms on subsecond wrist-worn accelerometry: a comparison using publicly available data sets”. *Journal for the Measurement of Physical Behaviour* 7.1.

Leroux, A., Cui, E., Smirnova, E., **Muschelli, J.**, Schrack, J. A., Crainiceanu, C. M., (2024). “NHANES 2011-2014: objective physical activity is the strongest predictor of all-cause mortality”. *Medicine and science in sports and exercise* 56.10, pp. 1926–1934.

Wright, C., Meng, Q., Breshock, M. R., Atta, L., Taub, M. A., Jager, L. R., **Muschelli, J.**, Hicks, S. C., (2024). "Open case studies: statistics and data science education through real-world applications". *Journal of Statistics and Data Science Education* 32.4, pp. 331–344.

2023 Savonen, C., Wright, C., Hoffman, A. M., **Muschelli, J.**, Cox, K., Tan, F. J., Leek, J. T., (2023). "Open-source tools for training resources–OTTR". *Journal of Statistics and Data Science Education* 31.1, pp. 57–65.

Zhao, C., Tapera, T. M., Bagautdinova, J., Bourque, J., Covitz, S., Gur, R. E., Gur, R. C., Larsen, B., Mehta, K., Meisler, S. L., Murtha, K., **Muschelli, J.**, Roalf, D. R., Sydnor, V. J., Valcarcel, A. M., Shinohara, R. T., Cieslak, M., Satterthwaite, T. D., (2023). "ModelArray: an R package for statistical analysis of fixel-wise data". *Neuroimage* 271, p. 120037.

2022 Karas, M., **Muschelli, J.**, Leroux, A., Urbanek, J. K., Wanigatunga, A. A., Bai, J., Crainiceanu, C. M., Schrack, J. A., (2022). "Comparison of accelerometry-based measures of physical activity: retrospective observational data analysis study". *JMIR mHealth and uHealth* 10.7, e38077.

Pham, D. D., **Muschelli, J.**, Mejia, A. F., (2022). "ciftiTools: a package for reading, writing, visualizing, and manipulating CIFTI files in R". *NeuroImage* 250, p. 118877.

Qi, G., Dutta, D., Leroux, A., Ray, D., **Muschelli, J.**, Crainiceanu, C., Chatterjee, N., (2022). "Genome-wide association studies of 27 accelerometry-derived physical activity measurements identified novel loci and genetic mechanisms". *Genetic epidemiology* 46.2, pp. 122–138.

Redd, A. D., Peetluk, L. S., Jarrett, B. A., Hanrahan, C., Schwartz, S., Rao, A., Jaffe, A. E., Peer, A. D., Jones, C. B., Lutz, C. S., McKee, C. D., Patel, E. U., Rosen, J. G., Desany, H. G., McKay, H. S., **Muschelli, J.**, Andersen, K. M., Link, M. A., Wada, N., Baral, P., Young, R., Boon, D., Grabowski, M. K., Gurley, E. S., the Novel Coronavirus Research Compendium Team, (2022). "Curating the evidence about COVID-19 for frontline public health and clinical care: the novel coronavirus research compendium". *Public Health Reports* 137.2, pp. 197–202. DOI: 10.1177/00333549211058732.

Sharrock, M. F., Mould, W. A., Hildreth, M., Ryu, E. P., Walborn, N., Awad, I. A., Hanley, D. F., **Muschelli, J.,** (2022). "Bayesian deep learning outperforms clinical trial estimators of intracerebral and intraventricular hemorrhage volume". *Journal of Neuroimaging* 32.5, pp. 968–976.

Smirnova, E., Mallow, C., **Muschelli, J.**, Shao, Y., Thiboutot, J., Lam, A., Rule, A. M., Crainiceanu, C., Yarmus, L., (2022). "Predictive performance of selected breath volatile organic carbon compounds in stage 1 lung cancer". *Translational Lung Cancer Research* 11.6, p. 1009.

2021 Bennett, T. D., Moffitt, R. A., Hajagos, J. G., Amor, B., Anand, A., Bissell, M. M., Bradwell, K. R., Bremer, C., Byrd, J. B., Denham, A., DeWitt, P. E., Gabriel, D., Garibaldi, B. T., Girvin, A. T., Guinney, J., Hill, E. L., Hong, S. S., Jimenez, H., Kavuluru, R., Kostka, K., Lehmann, H. P., Levitt, E., Mallipattu, S. K., Manna, A., McMurry, J. A., Morris, M., **Muschelli, J.**, Neumann, A. J., Palchuk, M. B., Pfaff, E. R., Qian, Z., Qureshi, N., Russell, S., Spratt, H., Walden, A., Williams, A. E., Wooldridge, J. T., Yoo, Y. J., Zhang, X. T., Zhu, R. L., Austin, C. P., Saltz, J. H., Gersing, K. R., Haendel, M. A., Chute, C. G., Consortium, N. C. C. C. N., (July 2021). "Clinical Characterization and Prediction of Clinical Severity of SARS-CoV-2 Infection Among US Adults Using Data From the US National COVID Cohort Collaborative". *JAMA Network*

Open 4.7, e2116901–e2116901. ISSN: 2574-3805. DOI: 10.1001/jamanetworkopen.2021.16901. eprint: https://jamanetwork.com/journals/jamanetworkopen/articlepdf/2781923/bennett_2021_oi_210506_1625605077.56265.pdf. URL: https://doi.org/10.1001/jamanetworkopen.2021.16901.

Bowring, M. G., Wang, Z., Xu, Y., Betz, J., **Muschelli, J.**, Garibaldi, B. T., Zeger, S. L., (2021). "Outcome-stratified analysis of biomarker trajectories for patients infected with severe acute respiratory syndrome coronavirus 2". *American journal of epidemiology* 190.10, pp. 2094–2106.

Broll, S., Urbanek, J., Buchanan, D., Chun, E., **Muschelli, J.**, Punjabi, N. M., Gaynanova, I., (2021). "Interpreting blood glucose data with R package iglu". *PloS one* 16.4, e0248560.

Garibaldi, B. T., Fiksel, J., **Muschelli, J.**, Robinson, M. L., Rouhizadeh, M., Perin, J., Schumock, G., Nagy, P., Gray, J. H., Malapati, H., Ghobadi-Krueger, M., Niessen, T. M., Kim, B. S., Hill, P. M., Ahmed, M. S., Dobkin, E. D., Blanding, R., Abele, J., Woods, B., Harkness, K., Thiemann, D. R., Bowring, M. G., Shah, A. B., Wang, M.-C., Bandeen-Roche, K., Rosen, A., Zeger, S. L., Gupta, A., (2021). "Patient trajectories among persons hospitalized for COVID-19: a cohort study". *Annals of Internal Medicine* 174.1, pp. 33–41.

Leroux, A., Xu, S., Kundu, P., **Muschelli, J.**, Smirnova, E., Chatterjee, N., Crainiceanu, C., (2021). "Quantifying the predictive performance of objectively measured physical activity on mortality in the UK Biobank". *The Journals of Gerontology: Series A* 76.8, pp. 1486–1494.

Scully, E. P., Schumock, G., Fu, M., Massaccesi, G., **Muschelli, J.**, Betz, J., Klein, E. Y., West, N. E., Robinson, M., Garibaldi, B. T., (2021). "Sex and gender differences in testing, hospital admission, clinical presentation, and drivers of severe outcomes from COVID-19". *Open forum infectious diseases*. Vol. 8. 9, ofab448.

Sharrock, M. F., Mould, W. A., Ali, H., Hildreth, M., Awad, I. A., Hanley, D. F., **Muschelli, J.,** (2021). "3D deep neural network segmentation of intracerebral hemorrhage: development and validation for clinical trials". *Neuroinformatics* 19.3, pp. 403–415.

Torbati, M. E., Minhas, D. S., Ahmad, G., O'Connor, E. E., **Muschelli, J.**, Laymon, C. M., Yang, Z., Cohen, A. D., Aizenstein, H. J., Klunk, W. E., (2021). "A multi-scanner neuroimaging data harmonization using RAVEL and ComBat". *Neuroimage* 245, p. 118703.

Tustison, N. J., Cook, P. A., Holbrook, A. J., Johnson, H. J., **Muschelli, J.**, Devenyi, G. A., Duda, J. T., Das, S. R., Cullen, N. C., Gillen, D. L., (2021). "The ANTsX ecosystem for quantitative biological and medical imaging". *Scientific reports* 11.1, pp. 1–13.

Wang, G., **Muschelli, J.**, Lindquist, M. A., (2021). "Moderated t-tests for group-level fMRI analysis". *NeuroImage* 237, p. 118141.

Wongvibulsin, S., Garibaldi, B. T., Antar, A. A. R., Wen, J., Wang, M.-C., Gupta, A., Bollinger, R., Xu, Y., Wang, K., Betz, J. F., **Muschelli, J.**, Bandeen-Roche, K., Zeger, S. L., Robinson, M. L., (2021). "Development of severe COVID-19 adaptive risk predictor (SCARP), a calculator to predict severe disease or death in hospitalized patients with COVID-19". *Annals of internal medicine* 174.6, pp. 777–785.

Wrobel, J., **Muschelli, J.**, Leroux, A., (2021). "Diurnal physical activity patterns across ages in a large UK based cohort: the UK Biobank study". *Sensors* 21.4, p. 1545.

2020 **Muschelli, J.** (2020). “A publicly available, high resolution, unbiased CT brain template”. *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*, pp. 358–366.

Ding, T, Cohen, A., O’Connor, E., Karim, H., Crainiceanu, A, **Muschelli, J**, Lopez, O, Klunk, W., Aizenstein, H., Krafty, R, (2020). “An improved algorithm of white matter hyperintensity detection in elderly adults”. *NeuroImage: Clinical* 25, p. 102151.

Gherman, A., **Muschelli, J.**, Caffo, B., Crainiceanu, C., (2020). “Rxnat: an open-source R package for XNAT-based repositories”. *Frontiers in Neuroinformatics* 14.

Hannawi, Y., **Muschelli, J.**, Mulder, M., Sharrock, M., Storm, C., Leithner, C., Crainiceanu, C. M., Stevens, R. D., (2020). “Postcardiac arrest neurological prognostication with quantitative regional cerebral densitometry”. *Resuscitation* 154, pp. 101–109.

Hansen, B. M., Ullman, N., **Muschelli, J.**, Norrving, B., Dlugash, R., Avadhani, R., Awad, I., Zuccarello, M., Ziai, W. C., Hanley, D. F., (2020). “Relationship of white matter lesions with intracerebral hemorrhage expansion and functional outcome: MISTIE II and CLEAR III”. *Neurocritical Care*, pp. 1–9. DOI: 10.1007/s12028-020-00916-4.

Kross, S., Leek, J. T., **Muschelli, J.,** (2020). “ari: The Automated R Instructor”. *The R Journal* 12.1, pp. 258–265. DOI: 10.32614/RJ-2020-023. URL: https://doi.org/10.32614/RJ-2020-023.

Minhas, D. S., Yang, Z., **Muschelli, J.**, Laymon, C. M., Mettenburg, J. M., Zammit, M. D., Johnson, S., Mathis, C. A., Cohen, A. D., Handen, B. L., (2020). “Statistical methods for processing neuroimaging data from two different sites with a Down syndrome population application”. *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*, pp. 367–379.

Rothstein, J. D., Caulfield, L. E., Broaddus-Shea, E. T., **Muschelli, J.**, Gilman, R. H., Winch, P. J., (2020). “‘The doctor said formula would help me’: health sector influences on use of infant formula in peri-urban Lima, Peru”. *Social Science & Medicine* 244.112324. DOI: https://doi.org/10.1016/j.socscimed.2019.05.029.

Ryan, S. M., Vestal, B., Maier, L. A., Carlson, N. E., **Muschelli, J.,** (2020). “Template creation for high-resolution computed tomography scans of the lung in R software”. *Academic radiology* 27.8, e204–e215.

Valcarcel, A. M., **Muschelli, J.**, Pham, D. L., Martin, M. L., Yushkevich, P., Brandstadter, R., Schindler, M. K., Patterson, K. R., Calabresi, P. A., Bakshi, R., Shinohara, R. T., (2020). “TAPAS: a thresholding approach for probability map automatic segmentation in multiple sclerosis”. *NeuroImage: Clinical*.

2019 **Muschelli, J.** (2019b). “ROC and AUC with a binary predictor: a potentially misleading metric”. *Journal of Classification*, pp. 1–13.

Muschelli, J. (2019a). “Recommendations for processing head CT data”. *Frontiers in Neuroinformatics* 13, p. 61. ISSN: 1662-5196. DOI: 10.3389/fninf.2019.00061. URL: https://www.frontiersin.org/article/10.3389/fninf.2019.00061.

Hadavand, A., **Muschelli, J.**, Leek, J., (2019). “Analysis of student behavior using the R package crsra”. *Journal of Learning Analytics* 6.2, pp. 140–152.

2018 **Muschelli, J.**, Sweeney, E., Crainiceanu, C. M., (2018). “freesurfer: connecting the Freesurfer software with R”. *F1000Research* 7.

Muschelli, J., Gherman, A., Fortin, J.-P., Avants, B., Whitcher, B., Clayden, J. D., Caffo, B. S., Crainiceanu, C. M., (2018). “Neuroconductor: an R platform for medical imaging analysis”. *Biostatistics*, kxx068. DOI: 10.1093/biostatistics/kxx068. eprint: /oup/backfile/content_public/journal/biostatistics/pap/10.1093_biostatistics_kxx068/1/kxx068.pdf. URL: +http://dx.doi.org/10.1093/biostatistics/kxx068.

Commowick, O., Istace, A., Kain, M., Laurent, B., Leray, F., Simon, M., Pop, S. C., Girard, P., Ameli, R., Ferré, J.-C., Kerbrat, A., Tourdias, T., Cervenansky, F., Glatard, T., Beaumont, J., Doyle, S., Forbes, F., Knight, J., Khademi, A., Mahbod, A., Wang, C., Mckinley, R., Wagner, F., **Muschelli, J.**, Sweeney, E., Roura, E., Lladó, X., Santos, M. M., Santos, W. P., Silva-Filho, A. G., Tomas-Fernandez, X., Urien, H., Bloch, I., Valverde, S., Cabezas, M., Vera-Olmos, F. J., Malpica, N., Guttmann, C., Vukusic, S., Edan, G., Dojat, M., Styner, M., Warfield, S. K., Cotton, F., Barillot, C., (2018). “Objective evaluation of multiple sclerosis lesion segmentation using a data management and processing infrastructure”. *Scientific Reports* 8, p. 13650.

Smith, C. H., Turbitt, E., **Muschelli, J.**, Leonard, L., Lewis, K. L., Freedman, B., Muratori, M., Biesecker, B. B., (2018). “Feasibility of coping effectiveness training for caregivers of children with autism spectrum disorder: a genetic counseling intervention”. *Journal of genetic counseling* 27.1, pp. 252–262.

Valcarcel, A., **Muschelli, J**, Crainiceanu, C, Pham, D., Calabresi, P., Bakshi, R, Shinohara, R., (2018c). “TAPAS: threshold adjustment to probability map automatic segmentations”. *MULTIPLE SCLEROSIS JOURNAL*. Vol. 24, pp. 629–630.

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Valcarcel, A. M., Linn, K. A., Khalid, F., Vandekar, S. N., Tauhid, S., Satterthwaite, T. D., **Muschelli, J.**, Bakshi, R., Shinohara, R. T., (2018a). “MIMoSA: an approach to automatically segment T2 hyperintense and T1 hypointense lesions in multiple sclerosis”. *International MICCAI Brainlesion Workshop*, pp. 47–56.

2017 **Muschelli, J.**, Sweeney, E. M., Ullman, N. L., Vespa, P., Hanley, D. F., Crainiceanu, C. M., (2017). “PItcHPERFeCT: primary intracranial hemorrhage probability estimation using random forests on CT”. *NeuroImage: Clinical* 14, pp. 379 –390. ISSN: 2213-1582. DOI: http://dx.doi.org/10.1016/j.nicl.2017.02.007. URL: http://www.sciencedirect.com/science/article/pii/S2213158217300414.

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2013 Mould, W. A., Carhuapoma, J. R., **Muschelli, J.**, Lane, K., Morgan, T. C., McBee, N. A., Bistran-Hall, A. J., Ullman, N. L., Vespa, P., Martin, N. A., Awad, I., Zuccarello, M., Hanley, D. F., (2013a). “Minimally invasive surgery plus recombinant tissue-type plasminogen activator for intracerebral hemorrhage evacuation decreases perihematomal edema”. *Stroke* 44.3, pp. 627–634.

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2012 Bundy, D. G., **Muschelli, J.**, Clemens, G. D., Strouse, J. J., Thompson, R. E., Casella, J. F., Miller, M. R., (2012). “Ambulatory care connections of medicaid-insured children with sickle cell disease”. *Pediatric Blood & Cancer* 59.5, pp. 888–894.

Eloyan, A., **Muschelli, J.**, Nebel, M. B., Liu, H., Han, F., Zhao, T., Barber, A. D., Joel, S., Pekar, J. J., Mostofsky, S. H., (2012). “Automated diagnoses of attention deficit hyperactive disorder using magnetic resonance imaging”. *Frontiers in Systems Neuroscience* 6, p. 6.

Hinson, H. E., Melnychuk, E., **Muschelli, J.**, Hanley, D. F., Awad, I. A., Ziai, W. C., (2012). “Drainage efficiency with dual versus single catheters in severe intraventricular hemorrhage”. *Neurocritical Care* 16.3, pp. 399–405.

Jaffe, J., Melnychuk, E., **Muschelli, J.**, Ziai, W., Morgan, T., Hanley, D. F., Awad, I. A., (2012). "Ventricular catheter location and the clearance of intraventricular hemorrhage". *Neurosurgery* 70.5, pp. 1258–1264.

Webb, A. J., Ullman, N. L., Mann, S., **Muschelli, J.**, Awad, I. A., Hanley, D. F., (2012). "Resolution of intraventricular hemorrhage varies by ventricular region and dose of intraventricular thrombolytic the clot lysis: evaluating accelerated resolution of IVH (CLEAR IVH) program". *Stroke* 43.6, pp. 1666–1668.

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2011 Newell, D. W., Shah, M. M., Wilcox, R., Hansmann, D. R., Melnychuk, E., **Muschelli, J.**, Hanley, D. F., (2011). "Minimally invasive evacuation of spontaneous intracerebral hemorrhage using sonothrombolysis". *Journal of Neurosurgery* 115.3, pp. 592–601.

Niedner, M. F., Huskins, W. C., Colantuoni, E., **Muschelli, J.**, Harris, J. M., Rice, T. B., Brilli, R. J., Miller, M. R., (2011). "Epidemiology of central line-associated bloodstream infections in the pediatric intensive care unit". *Infection Control* 32.12, pp. 1200–1208.

Working Groups

2014–Present **Biostatistics Structural Imaging Research Group**.

2014–Present **Penn Statistical Imaging and Visualization Endeavor (PennSIVE) Working Group**, *University of Pennsylvania, Department of Biostatistics and Epidemiology*.

2009–Present **Statistical Methods and Applications for Research in Technology (SMART) Working Group**, *Johns Hopkins University, Department of Biostatistics*.

2012–2016 **Epidemiology and Biostatistics of Aging (EBA) Training Program Meeting**, *Johns Hopkins University, Center on Aging and Health*.

Talks and Presentations

Invited Seminars

NHANES Accelerometry and Derivatives ,
Rollins School of Public Health - Emory University, Atlanta, GA, Invited Talk.

Neuroconductor: An R Platform for Medical Imaging Analysis ,
University of Arkansas for Medical Sciences (UAMS), Little Rock, AK, Invited Seminar.

Scientific Meetings

2025 **Processing accelerometer data in R**,
Emerging Statistical Methods for Digital Health Data, Tutorial.

2020 **R Package Development**,
Eastern North American Region (ENAR), (Online due to COVID19), Tutorial.

2019 **Potential Batch Effects and Biases in the UK Biobank Accelerometer Data**,
ENAR, Philadephia, PA, Talk.

2018 **Neuroconductor and Reproducibility: Imaging in R**,
Joint Statistical Meeting (JSM), Vancouver, BC, Canada, Talk.

My First Exposure to Accelerometer Data was for 100000 People from UK Biobank,
11th International Conference of the ERCIM WG on Computational and Methodological Statistics, Pisa, Italy, Talk.

Submitting to CRAN and Continuous Integration,
R Package Hackathon at Statistics in Imaging Conference, Philadelphia, PA, Talk and Hackathon Co-organizer.

Imaging Statistics in R,
Statistics in Imaging Conference, Philadelphia, PA, Talk.

Neuroimaging Analysis within R,
Eastern North American Region (ENAR), Atlanta, GA, Short Course.

Robust Lesion Segmentation on MRI of Patients with Multiple Sclerosis,
Genentech, South San Francisco, CA, Talk.

2017 **Neuroconductor: A Framework for a Framework for Reproducible Neuroimaging Analysis in R**,
Eastern North American Region (ENAR), Washington, DC, Poster.

Creating Interactive Graphics,
Joint Statistical Meeting (JSM), Baltimore, MD, Discussant and Organizer.

Neuroconductor: A Neuroimaging Analysis Project in R,
University of Mississippi Medical Center (UMMC), Jackson, MS, Invited Seminar.

Creating Interactive Graphics,
Joint Statistical Meeting (JSM), Baltimore, MD, Discussant and Organizer.

2016 **Papayar: A Better Interactive Neuroimage Plotter in R**,
Joint Statistical Meeting (JSM), Chicago, IL, Talk.

Processing Neuroimaging Data in R: Capabilities,
Mathematical and Statistical Challenges in Neuroimaging Data Analysis, Banff, AB, Talk.

Processing fMRI Data in R,
SAMSI Challenges in Functional Connectivity Modeling and Analysis Workshop, Durham, NC, Talk.

2015 **Succeeding in Undergraduate: A Message to Top Students**,
Sun Valley High School, Aston, PA, Talk.

SuBGELS: Subtraction-Based Gadolinium-Enhancing Lesion Segmentation,
Hopkins Imaging Conference, Baltimore, MD, Poster.

Automated Intracerebral Hemorrhage Segmentation of CT Scans,
Joint Statistical Meeting (JSM), Seattle, WA, SPEED Talk and Poster.

PItcHPERFECT: Primary Intracerebral Hemorrhage Prediction Employing Regression and Features Extracted from CT,
Eastern North American Region (ENAR), Miami, FL, Poster.

Quantitative Localization and Predictive Performance of Intracranial Hemorrhage,
International Stroke Conference (ISC), Nashville, TN, Poster.

Validated Automatic Brain Extraction of Head CT Images,
Organization for Human Brain Mapping (OHBM), Honolulu, HI, Poster.

2014 **Validated Automatic Brain Extraction of Head CT Images**,
Hopkins Imaging Conference, Baltimore, MD, Talk and Poster.

Reduction of motion-related artifacts in resting state fMRI using aCompCor,
Hopkins Imaging Conference, Baltimore, MD, Poster.
Award: Top Poster

2013 **Visualizing Brain Imaging in Interactive 3D**,
ENAR, Orlando, FL, Talk.

2012 **Resting State Preprocessing and Motion Artifacts**,
Second Biennial Conference on Resting State, Madgeburg, Germany, Poster.

Effects of preprocessing on motion-induced artifacts in resting state fMRI,
Society for Neuroscience (SfN), New Orleans, LA, Poster.